3-Methylglutaconic Aciduria Type VIIA (MGCA7A): A Comprehensive Disease Characterization Report

Disease: 3-Methylglutaconic Aciduria Type VIIA **MONDO ID:** MONDO:0859237 · **OMIM:** #616271 · **ORPHA:** 445038 **Causal gene:** *CLPB* (SKD3; HGNC:18542; OMIM gene *616254; chr11q13.4) **Category:** Mendelian (mitochondrial disorder)

Summary

3-Methylglutaconic aciduria type VIIA (MGCA7A, also written MGCA7) is an **ultra-rare inherited mitochondrial disorder** caused by pathogenic variants in *CLPB*, the gene encoding **SKD3**, an ATP-dependent AAA+ protein **disaggregase/refoldase** that resides in the mitochondrial intermembrane space (IMS). Two independent 2015 exome/homozygosity-mapping studies established the gene–disease relationship, and the disorder now sits at the intersection of **inborn errors of metabolism, congenital neutropenia, and neurodevelopmental disease**. It is inherited chiefly as an **autosomal recessive** trait (biallelic *CLPB* variants), with a distinct **dominant de novo monoallelic** subset acting through a dominant-negative mechanism (PMID: 25597510, PMID: 25597511, PMID: 34140661).

The clinical hallmark is a **graded multisystem tetrad**: congenital **neutropenia** (with risk of progression to bone-marrow failure, myelodysplastic syndrome, and acute leukemia), **neurodevelopmental impairment** ranging from nonprogressive intellectual disability to prenatal encephalopathy with progressive cerebral/cerebellar atrophy, movement disorder and epilepsy, bilateral **cataracts**, and **3-methylglutaconic aciduria** detectable on urine organic-acid screening. Postpubertal female survivors frequently develop **premature ovarian insufficiency (POI)**. Disease severity spans a continuum from neonatal death to survival into adulthood, and this severity tracks with the degree of **residual CLPB (refoldase) activity** retained by the mutant protein.

Mechanistically, loss of CLPB disaggregase activity destabilizes IMS client proteins—most notably the anti-apoptotic protein **HAX1** (itself mutated in Kostmann severe congenital neutropenia)—linking the enzymatic lesion to neutropenia and apoptosis. There is **no curative therapy**; management is supportive (G-CSF for neutropenia, cataract surgery, antiepileptics, infection prophylaxis, and hematologic surveillance), and **prevention is entirely genetic** (counseling, carrier/cascade testing with a Greenlandic Inuit founder emphasis, and prenatal/preimplantation genetic testing). This report synthesizes 12 confirmed findings across 17 reviewed papers into a complete disease-knowledge-base entry organized by the requested domains.

Section 1 — Disease Information

MGCA7A is a Mendelian mitochondrial disorder defined at the molecular level by loss of function of the *CLPB*-encoded disaggregase. **Key identifiers:** MONDO:0859237 · OMIM phenotype #616271 · OMIM gene *616254 (*CLPB*) · Orphanet ORPHA:445038 · gene HGNC:18542 · UniProt Q9H078. It is classified among the **inborn errors of metabolism with 3-methylglutaconic aciduria as a discriminative feature** (PMID: 23296368).

Synonyms / alternative names: 3-methylglutaconic aciduria type VII; MGCA7; MGCA7A; CLPB deficiency; "MEGCANN" (Methylglutaconic aciduria, cataracts, neurologic involvement, neutropenia); CLPB-related mitochondrial disorder with cataract, neutropenia, epilepsy and methylglutaconic aciduria.

The evidence base is **aggregated disease-level** (OMIM/Orphanet curation, cohort studies, mechanistic biochemistry) supplemented by **individual patient reports/case series**; there is no large EHR-derived dataset for this ultra-rare condition.

Section 2 — Etiology

Disease causal factor: purely genetic. MGCA7A is a monogenic disorder of *CLPB*. There is **no environmental, infectious, or multifactorial cause.** Two mutational mechanisms operate:

- **Biallelic (autosomal recessive) loss-of-function** — the classical form.
- **Monoallelic de novo dominant-negative** missense variants — a distinct subset.

Genetic risk factors: the only risk factor is possession of pathogenic *CLPB* alleles. **Consanguinity** and **founder effects** (Greenlandic/Inuit) raise biallelic risk in defined populations. No modifier loci have been formally mapped, though variant class strongly modifies severity (see Section 4/11).

Environmental risk / protective factors: none identified; diet, toxins, lifestyle, age, and sex are not causal. POI affects only females (a sex-specific manifestation, not a risk factor for the disease itself).

Gene–environment interactions: none documented. The oxidation-sensitive Y272C mutation introduces an aberrant disulfide bond (PMID: 37041140), raising a theoretical (unproven) redox-environment interaction, but no clinical GxE effect is established.

Section 3 — Phenotypes

Across cohorts the phenotype comprises **congenital neutropenia** (which can progress to leukemia), a **neurological spectrum** from nonprogressive intellectual disability to prenatal encephalopathy with progressive brain atrophy, **movement disorder**, **cataracts**, and elevated **urinary 3-methylglutaconic acid**. The pivotal review states the disorder spans "*a neurological phenotype ranging from nonprogressive intellectual disability to prenatal encephalopathy with progressive brain atrophy, movement disorder, cataracts, 3-methylglutaconic aciduria, and neutropenia*" (PMID: 34140661).

Pronicka et al. (31 patients) stratified severity into **mild (n=4), moderate (n=13), severe (n=14)**; the severe subtype featured "*the neonatal absence of voluntary movements in combination with ventilator dependency and hyperexcitability*," making MGCA7A an important differential diagnosis of **hyperekplexia** (PMID: 28687938). Postpubertal female survivors commonly develop premature ovarian insufficiency: "*POI is a common feature of postpubertal female individuals with CLPB deficiency*" (PMID: 36074910).

Phenotype	Type	HPO term	Onset	Severity/progression	Frequency
Neutropenia	Lab abnormality	HP:0001875	Congenital	Variable; leukemia risk	Very frequent
Intellectual disability	Clinical sign	HP:0001249	Congenital/ early	Nonprogressive→severe	Frequent
Cerebral atrophy	Imaging	HP:0002059	Prenatal–early	Progressive	Frequent
Cerebellar atrophy	Imaging	HP:0001272	Prenatal–early	Progressive	Frequent
Cataract	Physical	HP:0000518	Congenital/ early	Stable (surgical)	Frequent
3-Methylglutaconic aciduria	Lab abnormality	HP:0003535	Congenital	Persistent	Obligate marker
Movement disorder	Clinical sign	HP:0100022	Early	Variable	Frequent
Seizures/epilepsy	Clinical sign	HP:0001250	Early	Variable	Frequent
Neonatal hyperexcitability	Clinical sign	HP:0032322	Neonatal	Severe subtype	Severe form
Premature ovarian insufficiency	Clinical sign	HP:0008209	Postpubertal	Progressive	Common (female survivors)
Respiratory insufficiency	Clinical sign	HP:0002093	Neonatal	Severe form	Severe subtype

Quality of life impact: severe neonatal disease causes ventilator dependency, absent voluntary movement, and early death (median 35 days in the severe group), with profound QoL loss; milder disease permits survival into adulthood (median age 17.5 years in the mild group) but with intellectual disability, visual impairment from cataracts, and (in females) infertility from POI (PMID: 28687938, PMID: 36074910). No formal EQ-5D/SF-36 data exist.

Section 4 — Genetic / Molecular Information

Causal gene: *CLPB* (caseinolytic peptidase B homolog; chr11q13.4; HGNC:18542; OMIM *616254), encoding the mitochondrial AAA+ ATPase/disaggregase **SKD3** (UniProt Q9H078). Wortmann et al. identified "*mutations in CLPB define a syndrome*" in **14 individuals from 9 families** (PMID: 25597510). Saunders et al. independently reported "*biallelic variants in the caseinolytic peptidase B homolog (CLPB). The causative variants included one missense variant, c.803C>T (p.Thr268Met), and two nonsense variants, c.961A>T (p.Lys321*) and c.1249C>T (p.Arg417*)*" (PMID: 25597511).

Variant types & classification: missense, nonsense, frameshift, and splicing variants are all reported (pathogenic/likely pathogenic per ACMG). Stop/stop (nonsense-mediated decay) and ankyrin/N-domain-disrupting alleles confer the greatest burden. The **dominant subset** comprises de novo monoallelic missense variants: "*we identified one of four different de novo monoallelic missense variants in CLPB. We show that these variants disturb refoldase and to a lesser extent ATPase activity of CLPB in a dominant-negative manner*" (PMID: 34140661).

Functional consequences: biallelic variants cause **loss of function** (markedly decreased CLPB protein in patient fibroblasts/liver); de novo monoallelic variants act by **dominant-negative** interference with the oligomeric disaggregase. **Allele frequency:** all causal alleles are rare/absent in gnomAD, consistent with an ultra-rare disorder; the Greenlandic founder allele is enriched locally.

Structural insights: the N-domain mutation Y272C "*forms an intramolecular disulfide bond with Cys267 and severely impairs SKD3*" (PMID: 37041140). SKD3 assembles into **nucleotide-stabilized dodecamers** (two hexameric rings) (PMID: 35247700).

Modifier genes / epigenetics / chromosomal abnormalities: no formal modifier loci, methylation signatures, or large structural/chromosomal changes have been described; disease is point-mutation/small-variant driven.

Section 5 — Environmental Information

Not applicable. MGCA7A is a monogenic disorder with **no environmental, lifestyle, or infectious contribution.** No toxins, radiation, occupational exposures, dietary factors, or pathogens are implicated in causation or triggering. (The 3-methylglutaconic acid elevation is an endogenous metabolic consequence, not an environmental exposure.)

Section 6 — Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

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1. Biallelic LoF (recessive) OR de novo monoallelic dominant-negative
   CLPB variant
        │  leads to
        ▼
2. Reduced/abolished SKD3 (CLPB) ATP-dependent DISAGGREGASE/REFOLDASE
   activity in the mitochondrial intermembrane space
   (ATPase activity may be preserved; REFOLDASE loss is the key defect)
        │  results in
        ▼
3. Failure to disaggregate/refold IMS client proteins
        │
        ├──► 3a. Reduced solubility & altered assembly of anti-apoptotic
        │        HAX1  →  granulocyte apoptosis / maturation arrest
        │              →  CONGENITAL NEUTROPENIA  →  infection risk;
        │                 MDS / acute leukemia risk
        │
        ├──► 3b. IMS proteostatic stress / mitochondrial membrane
        │        pathology  →  accumulation & urinary excretion of
        │        3-METHYLGLUTACONIC ACID (nonspecific mito marker)
        │
        ├──► 3c. Neuronal mitochondrial dysfunction  →  prenatal/early
        │        encephalopathy  →  PROGRESSIVE CEREBRAL & CEREBELLAR
        │        ATROPHY, intellectual disability, movement disorder,
        │        epilepsy  (inferred: direct neuronal apoptotic pathway
        │        not fully demonstrated)
        │
        ├──► 3d. Lens epithelial mitochondrial dysfunction  →  bilateral
        │        CATARACTS  (inferred)
        │
        └──► 3e. Ovarian (granulosa/oocyte) mitochondrial dysfunction  →
                 PREMATURE OVARIAN INSUFFICIENCY in postpubertal females
        │
        ▼
4. SEVERITY is set by RESIDUAL REFOLDASE ACTIVITY:
   stop/stop & NMD/ankyrin-disrupting variants → severe, neonatal death;
   missense with partial function → milder, survival into adulthood.
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Detail

Molecular pathway / protein function. SKD3 (human ClpB) couples **ATP hydrolysis to protein disaggregation/reactivation** in the IMS, regulated by PARL: "*A mitochondrial inner-membrane protease, PARL, removes an autoinhibitory peptide from Skd3 to greatly enhance disaggregase activity*" (PMID: 32573439). It localizes to the IMS—"*CLPB resides in the mitochondrial intermembrane space*" (PMID: 32866687)—where it forms ATP-dependent high-molecular-weight complexes/dodecamers (PMID: 35247700).

Biochemical/cellular consequence (the HAX1 bridge). Loss of disaggregase activity destabilizes clients: "*Human cells lacking Skd3 exhibit reduced solubility of various mitochondrial proteins, including anti-apoptotic Hax1. Importantly, Skd3 variants linked to 3-methylglutaconic aciduria... display diminished disaggregase activity (but not always reduced ATPase activity), which predicts disease severity*" (PMID: 32573439). Because **HAX1** is mutated in Kostmann congenital neutropenia, its destabilization provides a direct molecular route to the neutropenia phenotype and to dysregulated apoptosis. Complexome profiling confirms altered assembly: in patients "*multiple HAX1 peaks were observed at higher molecular masses comigrating with CLPB*" (PMID: 34140661).

Upstream vs downstream. The *CLPB* mutation and loss of refoldase activity are **upstream** (and encode severity). The HAX1/IMS-client destabilization is the **proximal effector** (best-demonstrated, explains neutropenia). The brain/lens/ovary manifestations are **downstream** and mechanistically **inferred** from shared mitochondrial proteostatic stress in high-energy tissues.

Cell types & processes involved: neutrophils/granulocytes (CL:0000775/CL:0000094), neurons and glia, lens epithelium, ovarian granulosa/oocytes.

Suggested GO terms: GO:0043335 (protein unfolding), GO:0042026 (protein refolding), GO:0016887 (ATP hydrolysis activity), GO:0005758 (mitochondrial intermembrane space), GO:0006915 (apoptotic process). **CHEBI:** CHEBI:28123 (3-methylglutaconic acid).

Metabolomics / 3-MGA-uria: 3-methylglutaconic aciduria is a **nonspecific marker of mitochondrial dysfunction**; its exact biochemical origin in CLPB disease is unknown (PMID: 23355087, PMID: 23296368). SKD3/CLPB is also upregulated in acute myeloid leukemia, where it contributes to venetoclax resistance (PMID: 35247700, PMID: 31048321).

Section 7 — Anatomical Structures Affected

Organ level (primary): central nervous system (progressive cerebral and cerebellar atrophy, movement disorder, epilepsy—"*Muscular hypo- and hypertonia, movement disorder and progressive brain atrophy are frequently reported*" PMID: 28687938); hematopoietic/bone marrow (granulocytic neutropenia with leukemia risk); ocular lens (bilateral cataracts); ovary (POI in female survivors). **Secondary involvement:** respiratory system and swallowing/feeding apparatus in severe neonates.

Body systems: nervous, hematologic/immune, ocular/visual, reproductive/endocrine.

Tissue/cell level: nervous tissue (neurons, cerebellar Purkinje/granule populations), granulocytic hematopoietic lineage, lens epithelium, ovarian granulosa cells. **Cell Ontology:** CL:0000775 (neutrophil), CL:0000094 (granulocyte).

Subcellular level: the **mitochondrial intermembrane space** (GO:0005758) is the primary locus—"*indicating that CLPB resides in the mitochondrial intermembrane space*" (PMID: 32866687).

UBERON terms: UBERON:0000955 (brain), UBERON:0002037 (cerebellum), UBERON:0002371 (bone marrow), UBERON:0000965 (lens), UBERON:0000992 (ovary). **Lateralization:** CNS atrophy and cataracts are typically **bilateral**.

Section 8 — Temporal Development

Onset: congenital, often **prenatal** in severe cases (reduced fetal movements). **Pattern:** subacute-to-chronic with a progressive neurological component.

Progression: "*The foetal and neonatal presentation mirrored the course of disease with respect to survival (current median age 17.5 years in the mild group, median age of death 35 days in the severe group), severity and age of onset of all findings evaluated*" (PMID: 28687938). The neurologic component is progressive (progressive brain atrophy); neutropenia and cataracts are congenital/early. Age of onset predicts rate: "*Early-onset disease (<12 months) correlated with more severe progression. Later onset often resulted in milder phenotypes*" (PMID: 41719910).

Course / duration: lifelong/chronic in survivors; severe disease is rapidly fatal in the neonatal period. **Critical period:** the neonatal window (respiratory/feeding support, G-CSF) is decisive for survival. No spontaneous remission; POI emerges postpubertally.

Section 9 — Inheritance and Population

Epidemiology: ultra-rare (ORPHA:445038; prevalence <1/1,000,000). Fewer than ~100 patients are reported worldwide; the largest cohort is 63 individuals (PMID: 41719910). No formal incidence estimate exists.

Inheritance: predominantly **autosomal recessive** (OMIM #616271; biallelic *CLPB*), with a distinct **autosomal dominant de novo** subset (PMID: 34140661).

Penetrance/expressivity: apparently complete penetrance in biallelic/dominant-negative genotypes but **highly variable expressivity**, graded by residual refoldase activity/variant class.

Founder effect / consanguinity: a **Greenlandic (Inuit) founder** population is documented —"*Four of the individuals were of Greenlandic descent, and one was North American, of Northern European and Asian descent*" (PMID: 25597511). Consanguinity contributes to biallelic cases in some populations. **Carrier frequency** is elevated in the founder population but not formally quantified globally.

Demographics: both sexes affected; POI is female-specific. No overall sex-ratio skew is established. **Genetic anticipation / germline mosaicism:** not a repeat-expansion disorder, so no anticipation; germline mosaicism is a theoretical counseling caveat for de novo cases.

Survival by zygosity:

Parameter	Biallelic	Monoallelic (de novo)
N (63-case cohort)	41	22
Mortality	66%	23%
Median age at death	6 months	2.4 years
4-year survival	50%	82%

"*Death occurred in 66% of biallelic and 23% of monoallelic individuals, with earlier median age at death in biallelic cases (6 months vs 2.4 years). Biallelic cases had significantly higher DBI scores and poorer survival (4-year survival: 50% vs 82%)*" (PMID: 41719910).

Section 10 — Diagnostics

Laboratory / biochemistry: urine **GC-MS** shows elevated 3-methylglutaconic (and 3-methylglutaric) acid—one neonate's "*urine 3-methylpentenedioic acid... increased to 17.53 µmol/L*" (PMID: 37906145). **CBC** demonstrates congenital neutropenia.

Imaging: brain **MRI** shows progressive cerebral/cerebellar atrophy and may show structural anomalies; **ophthalmologic exam** detects cataracts.

Genetic testing (definitive): molecular confirmation via **NGS**—whole-exome/whole-genome sequencing or targeted panels (congenital neutropenia / 3-MGA / mitochondrial disease)—identifying biallelic or de novo monoallelic *CLPB* variants: "*metabolic analysis suggested 3-methylglutaconic aciduria due to elevated 3-methylglutaconic acid and 3-methylglutaric acid. NGS has detected two compound heterozygous CLPB variants*" (PMID: 32219827). WES/WGS is first-line; single-gene *CLPB* testing applies in known founder families. Chromosomal microarray/karyotype/FISH/mtDNA/repeat-expansion testing are **not indicated** (no structural or mtDNA lesions).

Clinical criteria / differential diagnosis: 3-MGA-uria is the diagnostic key but is nonspecific —"*There is... a group of disorders with significantly and consistently increased 3-methylglutaconic acid excretion, where the 3-methylglutaconic aciduria is a hallmark of the phenotype and the key to diagnosis*" (PMID: 23296368). Differentials with discriminative 3-MGA-uria include **TAZ (Barth syndrome), SERAC1 (MEGDEL), OPA3 (Costeff), DNAJC19 (DCMA), and TMEM70** (PMID: 23355087); among congenital neutropenia syndromes, CLPB disease must be distinguished from HAX1 (Kostmann), ELANE, G6PC3 and others (PMID: 38589208). Severe neonatal CLPB disease is an important differential of **hyperekplexia** (PMID: 28687938).

Omics diagnostics: RNAseq/quantitative proteomics/complexome profiling confirm causation in difficult cases (PMID: 36074910, PMID: 34140661). **Screening:** biochemical NBS for 3-MGA is not standard; carrier/cascade testing applies in founder families.

Suggested LOINC/ontology: urine organic acids (GC-MS); HP:0003535 (3-MGA-uria); HP:0001875 (neutropenia).

Section 11 — Outcome / Prognosis

Survival/mortality: strongly genotype-dependent. Biallelic disease carries 66% mortality (median death 6 months; 4-year survival 50%) versus 23% mortality for monoallelic disease (median death 2.4 years; 4-year survival 82%) (PMID: 41719910). The Pronicka cohort spanned "*current median age 17.5 years in the mild group, median age of death 35 days in the severe group*" (PMID: 28687938).

Morbidity/disability: survivors face intellectual disability, movement disorder, epilepsy, visual impairment (cataracts), recurrent infection (neutropenia), and—in females—infertility from POI. **Complications:** life-threatening infections; progression to **bone-marrow failure, MDS, and acute leukemia** (PMID: 38589208); respiratory failure in severe neonates.

Prognostic factors: variant class/structural prediction is the strongest—"*Structural predictions—particularly variants causing nonsense-mediated decay or ankyrin domain disruption—were stronger survival predictors than zygosity or age of onset*" (PMID: 41719910). Residual **disaggregase activity predicts disease severity** (PMID: 32573439). Early onset (<12 months) predicts worse course. No formal QoL metrics are published.

Section 12 — Treatment

No disease-modifying therapy exists; management is supportive/symptomatic.

Pharmacotherapy: granulocyte colony-stimulating factor (G-CSF) corrects neutropenia—"*Administration of granulocyte colony-stimulating factor successfully maintained neutrophil counts above 0.5 × 10^9/L and significantly reduced the frequency of respiratory tract infections*" (PMID: 41560087). Antiepileptics manage seizures; symptomatic agents address the movement disorder. **Pharmacogenomics:** none established.

Surgical: cataract extraction for visual rehabilitation. **Supportive/rehabilitative:** neonatal respiratory and feeding support in severe disease; developmental, physical, occupational, and speech therapy; endocrine/fertility care (hormone replacement) for POI (PMID: 36074910).

Caveat on G-CSF: residual malignancy risk persists—"*While some patients benefit from granulocyte colony-stimulating factor treatment, they may still face an increased risk of bone marrow failure, myelodysplastic syndrome and acute leukaemia*"—mandating hematologic

surveillance (PMID: 38589208). Hematopoietic stem cell transplantation is considered in severe-congenital-neutropenia disorders but is not an established cure for the multisystem CLPB phenotype.

Advanced/experimental therapeutics: no approved gene, cell, or RNA-based therapy; no disease-specific clinical trials with NCT identifiers were identified. **Suggested NCIT terms:** C1725 (G-CSF), C15329 (Cataract Surgery), C264 (Anticonvulsant Agent), C15431 (Supportive Care), C15431 (symptom management).

Section 13 — Prevention

Because MGCA7A is a monogenic *CLPB* disorder with **no environmental cause**, prevention is **entirely genetic**:

- **Primary/genetic prevention:** genetic counseling with recurrence-risk assessment (**25%** for carrier couples in AR families; low—but non-zero via germline mosaicism—for de novo dominant cases); **carrier testing** (targeted founder-variant testing in the Greenlandic/Inuit population); **cascade testing** of relatives; and reproductive options including **prenatal diagnosis** and **preimplantation genetic testing (PGT)** once familial variants are known (PMID: 25597511, PMID: 34140661).
- **Secondary prevention:** early diagnosis via urinary organic-acid screening + NGS and prompt supportive care (PMID: 41560087).
- **Tertiary prevention:** infection prophylaxis and **leukemia/MDS surveillance** (PMID: 38589208).

Immunization / public-health / environmental interventions: not applicable (no infectious or environmental etiology).

Section 14 — Other Species / Natural Disease

Taxonomy / orthologs: *CLPB* is evolutionarily conserved; a functional ortholog exists in **zebrafish (*Danio rerio*, NCBI Taxon 7955)** (used as a disease model, below). Bacterial/yeast ClpB/Hsp104 family members are the ancestral disaggregases, underscoring deep conservation of the disaggregase mechanism.

Naturally occurring disease in other species: no naturally occurring CLPB-deficiency disease has been catalogued in companion animals or wildlife (no OMIA entry identified). **Zoonotic potential / cross-species transmission:** not applicable (non-infectious genetic disorder).

Comparative biology: the conservation of the disaggregase from bacteria/yeast (Hsp104) through zebrafish to humans makes the mechanism broadly tractable, but no natural animal disease model is reported.

Section 15 — Model Organisms

Zebrafish (*Danio rerio*) morphant is the principal in vivo model: "*Suppression of clpb in zebrafish embryos induced a central nervous system phenotype that was consistent with cerebellar and cerebral atrophy that could be rescued by wild-type, but not mutant, human CLPB mRNA*" (PMID: 25597510). This both **recapitulates the CNS atrophy** and provides a **variant-pathogenicity rescue assay**. Its limitation is that it models mainly the neurological component, not the full multisystem syndrome.

In vitro / cellular models: - Recombinant **SKD3 ATPase/disaggregase activity assays** (PMID: 32573439, PMID: 35247700). - **Patient fibroblasts and liver** showing markedly decreased CLPB protein (PMID: 25597511). - **HEK-293/BT-549** cell fractionation establishing IMS localization and ATP-dependent complexes (PMID: 32866687). - **CLPB-knockout human cell lines** demonstrating reduced solubility of mitochondrial proteins including HAX1 (PMID: 32573439).

No published mouse knock-in recapitulating the human genotype–phenotype spectrum was identified—a key gap. **Resources:** ZFIN (zebrafish), Cellosaurus (HEK-293/BT-549).

Mechanistic Model / Interpretation (Synthesis)

The pathophysiology is best understood as a **single enzymatic lesion radiating into tissue-specific failures**. The proximal, best-demonstrated arm is the **CLPB→HAX1→neutropenia** axis: loss of the IMS disaggregase renders anti-apoptotic HAX1 insoluble and mis-assembled, phenocopying the HAX1-mutant Kostmann neutropenia. The remaining arms (brain atrophy, cataract, POI) are **inferred** to arise from generalized IMS proteostatic stress in highly mitochondria-dependent tissues. Severity is **encoded at the genotype**: variants that abolish refoldase activity (NMD/stop-stop, ankyrin-domain disruption) produce lethal neonatal disease,

whereas partially functional missense alleles allow survival into adulthood. Critically, **disaggregase (refoldase) activity—not ATPase activity—is the quantitative determinant of severity**, which reframes the protein's function (refolding, not merely ATP hydrolysis) as the therapeutically relevant target.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Supports
25597510	*CLPB mutations cause 3-MGA, brain atrophy, neutropenia, cataracts*	Human genetics + zebrafish	Gene discovery; zebrafish CNS model
25597511	*CLPB variants w/ cataract, neutropenia, epilepsy, MGA*	Human genetics	Independent confirmation; Greenlandic founder; variants
34140661	*Neutropenia & ID are hallmarks of biallelic & de novo CLPB deficiency*	Human genetics + complexome	Dominant-negative subset; HAX1 assembly
32573439	*Skd3 is a potent mito disaggregase inactivated by MGA mutations*	In vitro + human KO cells	PARL activation; HAX1 solubility; severity predictor
32866687	*CLPB forms ATP-dependent complexes in the IMS*	Cell biology	Subcellular localization
35247700	*SKD3 assembles into nucleotide-stabilized dodecamers*	Structural	Double-hexamer architecture; AML link
37041140	*Structural basis of impaired disaggregase in oxidation-sensitive SKD3 mutant*	Structural	Y272C/Cys267 disulfide mechanism
28687938	*Scoring system predicting clinical course (31 patients)*	Clinical cohort	Severity stratification; survival; temporal course
41719910	*Genotype to outcome: zygosity-specific insights (63 cases)*	Clinical cohort	Survival by zygosity; structural predictors
36074910	*Premature ovarian insufficiency in CLPB deficiency*	Clinical + multi-omics	POI; RNAseq/proteomics validation
41560087	*CLPB deficiency-associated congenital neutropenia (case report)*	Case report	G-CSF efficacy
38589208	*Congenital neutropenia syndromes: differential diagnosis*	Review	Residual leukemia/MDS risk; differential

PMID	Title (abbrev.)	Evidence type	Supports
32219827	*Clinical/genetic analysis of an infant with MGCA7*	Case report	Biochemical + NGS diagnosis
37906145	*Child with MGCA7 due to novel CLPB variants*	Case report	Urinary 3-MGA hallmark
23296368	*Inborn errors with 3-MGA-uria: classification & nomenclature*	Review	3-MGA-uria as diagnostic key
23355087	*3-MGA-uria: lessons from 50 genes and 977 patients*	Large review	Differential (TAZ, SERAC1, OPA3, DNAJC19, TMEM70)
31048321	*Targeting mitochondrial structure sensitizes AML to venetoclax*	Functional genomics	Context — CLPB/mito chaperone in AML

Evidence-source breakdown: Human clinical genetics and cohorts provide the strongest evidence for gene–disease relationship, phenotype, and prognosis. Mechanistic claims rest on in vitro biochemistry and human knockout-cell/complexome studies (strong for the HAX1 branch). Animal-model evidence is limited to the zebrafish morphant.

Limitations and Knowledge Gaps

1. **Rarity limits statistics.** Fewer than ~100 patients worldwide; largest cohort 63. Prevalence/incidence, penetrance, and expressivity are described qualitatively, not with confidence intervals.
2. **Mechanistic gaps for non-hematologic branches.** The HAX1–neutropenia link is biochemically demonstrated; the pathways to brain atrophy, cataract, and POI are **inferred** from shared mitochondrial stress, not directly shown. Specific IMS clients and apoptotic effectors in neurons, lens, and ovary are undefined.
3. **Origin of 3-MGA-uria unknown.** As in other mitochondrial 3-MGA-urias (except AUH), the biochemical origin of the excreted acid is unestablished; it is a nonspecific marker, not a direct CLPB readout.
4. **No disease-modifying therapy or validated progression biomarker.** G-CSF addresses only neutropenia and does not abolish malignancy risk; variant class/structural prediction is the only deployed prognostic tool.

5. **Limited animal models.** The zebrafish morphant captures CNS atrophy but not the full syndrome; no mouse knock-in recapitulating the human spectrum was identified.
6. **Modifier genes and epigenetics** underlying the wide severity range are essentially unstudied; founder data outside Greenland/Inuit are sparse.

Proposed Follow-up Experiments / Actions

1. **Generate a mouse (or iPSC-organoid) allelic series** spanning null, hypomorphic missense, and dominant-negative *CLPB* variants to test the "residual refoldase activity sets severity" model in vivo and to model the brain/lens/ovary branches directly.
2. **Define the IMS client interactome** by proximity-labeling (BioID/APEX) proteomics in CLPB-knockout vs. rescued human neurons, lens epithelium, and granulosa cells—to find tissue-specific aggregation-prone clients beyond HAX1.
3. **Develop a quantitative variant-function assay** (recombinant SKD3 disaggregase activity ± PARL processing) as a clinical-grade functional classifier to resolve VUS and refine prognosis (PMID: 41719910).
4. **Establish a prospective natural-history registry** with standardized neuro-imaging, CBC, ophthalmologic, and endocrine endpoints to generate confidence-bounded prevalence, penetrance, and progression data and to support trial design.
5. **Test chaperone/proteostasis-restoring or HAX1-stabilizing strategies** (e.g., molecules enhancing residual disaggregase function) in patient-derived cells as disease-modifying proof-of-concept.
6. **Pharmacovigilance for leukemia/MDS** in G-CSF-treated patients via longitudinal bone-marrow surveillance cohorts to quantify malignant-transformation hazard specific to CLPB deficiency.
7. **Expand founder-variant carrier screening** in Greenlandic/Inuit populations and investigate additional geographic founder alleles to inform targeted reproductive genetic services.

Report compiled from 12 confirmed findings across 17 reviewed primary and review articles. Evidence sources are distinguished as human clinical genetics/cohorts, in vitro biochemistry, human knockout-cell and complexome studies, structural biology, and a zebrafish morphant model. All mechanistic and clinical claims are cited to PMID with verbatim abstract support where available.
